

Lab 03

PMDC with Back-Emf Speed Sensing

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CSE 4355- Section 001

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Code Explanation

```
// Compute RPM from Optical Sensor
float OpticalSensorRPM(uint32_t EdgeCount, float SampleMS, uint32_t pulsesPerRevolution)
{
    // Calculate # Edges / Second
    // float EdgesPerSec = ((float)EdgeCount * 1000.0f);

    // Calculate Revolutions / Second
    // float RevsPerSec = EdgesPerSec / (float)pulsesPerRevolution;    // 32 Pulses / Revolution?

    float RevsPerSec = EdgeCount / (float)pulsesPerRevolution;
    // Calculate RPM
    float RPM = RevsPerSec * 60.0f ;                                // RevsPerSec * 60 s = RPM

    return RPM;
}

// Compute RPM from Back EMF
float BemfRPM(float BEMFVoltage)
{
    // Placeholder Coefficient (Replace with Measurements) ^^^^^^
    return (RPM_PER_VOLT * (BEMFVoltage - VOLTAGE_OFFSET));
}
```

```
// Main Loop
while (1)
{
    uint32_t now = MilliS();

    // Frequency Sampling Window
    if (now - WindowStart >= FREQ_SAMPLE_MS)
    {
        uint32_t edges = OpticalSensorEdgeCount - StartingCount;
        float rpm_freq = OpticalSensorRPM(edges, (float)FREQ_SAMPLE_MS, pulsesPerRevolution);

        // Read Last Back EMF Voltage saved by ADC interrupt routine (call CalculateBEMFVoltage() if needed)
        float Bemf_Volts = CurrentBEMFVoltage;
        float Bemf_RPM = BemfRPM(Bemf_Volts);

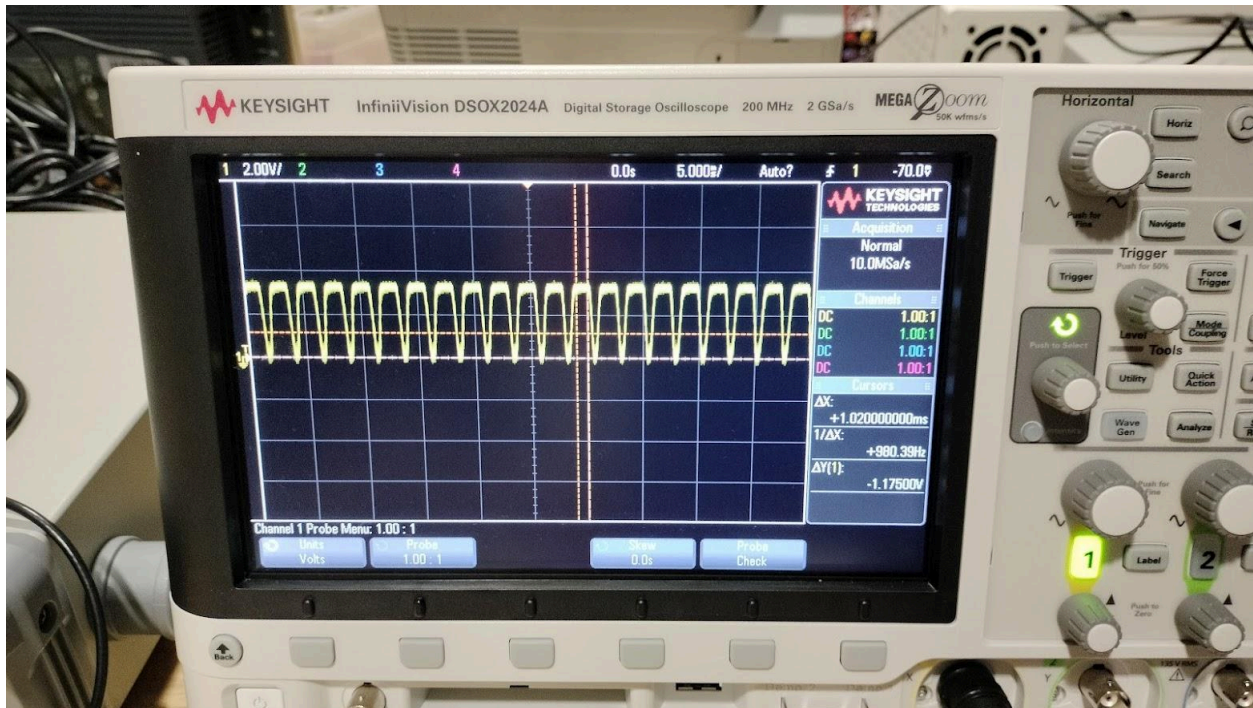
        // Print Readings
        putsUart0("\033[2J\033[H");
        putsUart0("\r\n--- Motor PWM + Back-EMF Demo ---\r\n");

        putsUart0("FreqEdges=\t");
        toAsciiDecimal(UARTBuff, edges);
        putsUart0(UARTBuff);
        putsUart0("\r\n");

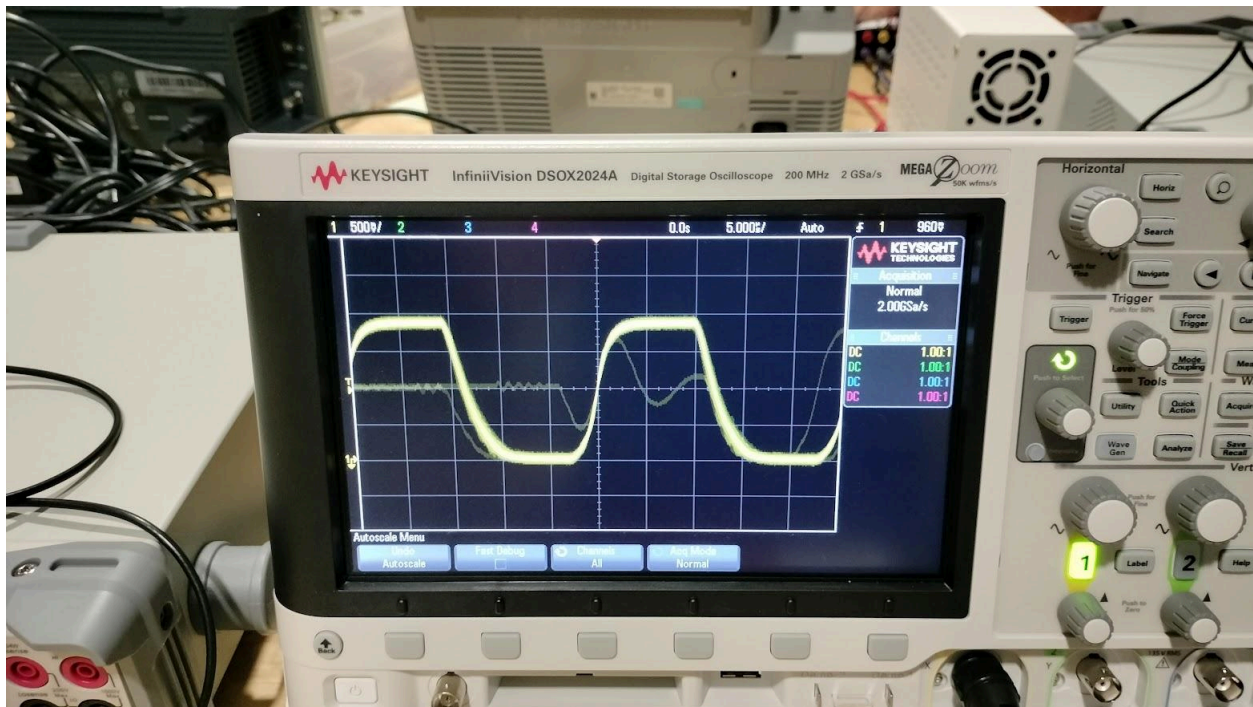
        putsUart0("RPM(freq)=\t");
        toAsciiDecimal(UARTBuff, (int32_t)frequency);
        putsUart0(UARTBuff);
        putsUart0(".");
        //int fractional = (int)(frequency * 100) % 100;
        int fractional = frequency;
        toAsciiDecimal(UARTBuff, fractional);
    }
}
```

Oscilloscope Pictures

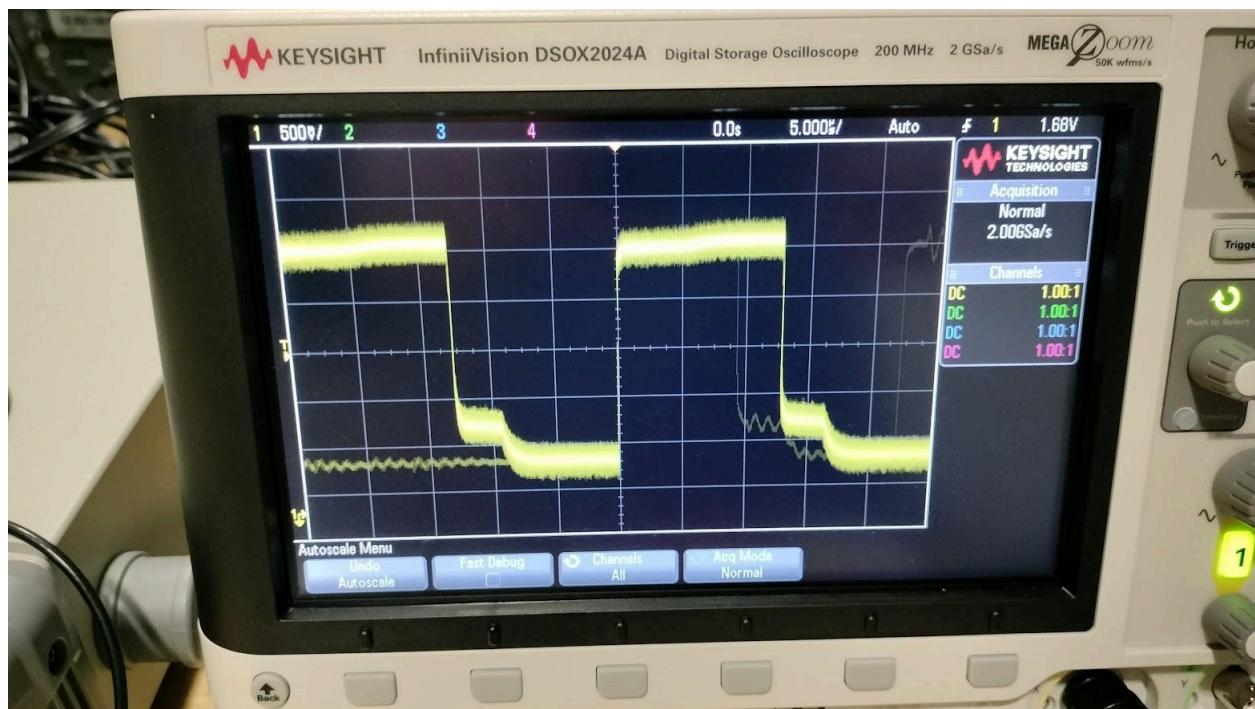
Optical Sensor



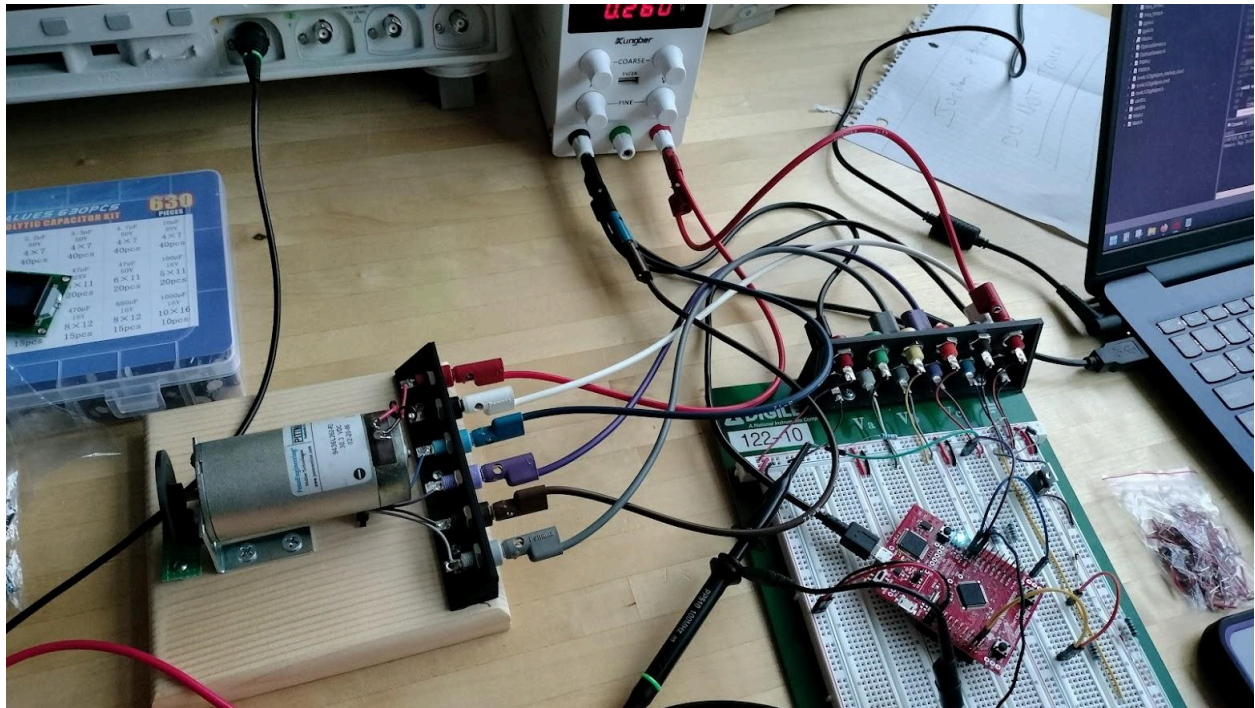
ADC



PWM



Hardware Pictures



Readings

Duty Cycle	RPM(freq)	ADC	RPM(bemf)
0%	0	10.8	0
10%	358	9.9	167
20%	637	9.5	424
30%	972	8.7	758
40%	1249	7.2	1038
50%	1609	5.2	1396
60%	1923	4.1	1717
70%	2270	3.5	1985
80%	2582	2.3	2358
90%	2905	0.9	2689
100%	3229	0	2979